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EDUCATION

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| ★ The University of Toronto | Toronto, ON, Canada | Aug 2017 – Dec 2022 (Expected) |
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- **Major: Mechanical Engineering;**
Minor: Robotics and Engineering Business in Rotman Commerce
 - **Technical Elective Coursework:** Mechatronics; Control System; Robotics and Manipulator; Embedded Systems; Computer Architecture; State Estimation; Machine Learning; OpenCV; Robot Operating System; Localization and Mapping; Path Planning; Linux; Bioengineering

RESEARCH EXPERIENCE

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| ★ Space and Terrestrial Autonomous Robotic Systems Lab | University of Toronto – Prof. Jonathon Kelly | Sep 2021 - Current |
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- Developed novel algorithms for 2D-radar-to-2D-radar and 3D-radar-to-camera extrinsic calibrations based on the radar/camera ego velocity estimation and rigid body motion. New Algorithm allow sensors to be calibrated online without assistance of external targets.
 - Implemented different filtering methods like RANSAC, Sliding-Window and Hampel filter for noisy radar data
 - Calibrated the camera intrinsic parameters via Kalibr, rectified the data using mage_undistort and SLAM pose estimation comparison using ORB_SLAM on ROS
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| ★ Advanced Controls Research Lab | University of Illinois – Prof. Naira Hovakimyan | July 2021 - Dec 2021 |
|----------------------------------|---|----------------------|
- Developed a novel lightweight chassis design for a unicycle robot via generative design
 - Researched on the 2DOF unicycle robot balance controls and maneuvering control using reactional wheels. Completed real-life testings and simulations on MATLAB, Simulink, and Simscape

EMPLOYMENT

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|---------------------------|----------------|----------------------|
| ★ BOE Technology Group Co | Beijing, China | May 2021 - July 2021 |
|---------------------------|----------------|----------------------|
- Software Engineer Intern**
- Developed and trained two CNN neural networks using TensorFlow and PyTorch that distinguish people's age, gender and clothing
 - Carried out web crawlers and extracted over 3000 useful video and picture data online. Doubled the usable data for machine training. Overall accuracy increased from 64% to 89%
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| ★ China State Shipbuilding Cooperation | Shanghai, China | Sep 2020 - April 2020 |
|--|-----------------|-----------------------|
- Mechanical Engineer Intern**
- Assisted designing the HVAC system layout of an upcoming 12-deck cruise ship using CADMATIC and Bentley
 - Designed and filed a patent for a high-pressure spray nozzle and its associated fire pipeline system. The resulted CFD simulation achieved using 1/5 volume of water compared with existing design, and can cover the fire in 1/3 of the time in simulation
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| ★ Autodesk | Toronto, ON, Canada | Sep 2019 - June 2020 |
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- Student Ambassador**
- Established Fusion Design Association Club and served as club president. Helped the company trained over 30 students on using Fusion 360's solid modeling, sculpture, generative design and sheet modeling features
 - Setup an annual 3D-printed glider competition and designed the glider launching system for the competition

LANGUAGES AND TECHNOLOGIES

- C; C++; Python; MATLAB; Simulink; Simscape; C; Linux; ROS(Robot Operating System); Numpy; PyTorch; Panda and TensorFlow
- SolidWorks; Altium Designer; KiCAD; Eagle; Fusion 360; AutoCAD; LTspice; Ansys

PUBLICATIONS

- "Spatiotemporal Calibration of 3D mm-Wavelength Radar-Camera Pairs" on IEEE Transactions on Robotics, Oct. 2022 as the second author
- "Generative Design for Self-balancing Unicycle Robot in Additive Manufacturing" on ACAIB 2022 as the first author
- "Weakly Supervised Semantic and Attentive Data Mixing Augmentation for Fine-Grained Visual Categorization" in IEEE Access, vol. 10, pp. 35814-35823, 2022 as the second author